REV2026 Course Code: 2049 Semester II Lab Course

Electronic Devices and Applications Lab

Hands-on experience in constructing and testing rectifiers, filters, clippers, clampers, and BJT characteristics.

Download Notes Print Lesson

Official Source Declaration

This lesson is strictly based on the official **SITTTR Kerala Revision 2026** syllabus for Course Code **2049**.

Official Source Note:

The syllabus PDF (Page 4) contains a slight numerical inconsistency in the Assessment Methodology table. The "Converted to" marks for CIE (Attendance 5 + CE 30 + CA1 15 + CA2 15 + OEE 10) sum to

75

, whereas the course header and total marks row state

60

. This lesson presents the data exactly as found in the official document.

Course Dashboard

Total Contact Hours

45

3 Hours/Week (Practical)

Credits

1.5

Lab Category

Assessment

CIE: 60 | ESE: 40

Total: 100 Marks

Pre-requisites

- **1041:** Elementary Concepts of Electronics (Sem I)
- **2042:** Electronic Devices and Applications (Sem II Theory)

How to Study this Lesson

This lab-oriented lesson is designed to bridge theory and practice. Follow these steps:

1. **Pre-Lab:** Read the theoretical principles in the module sections before attending the lab.
2. **Simulation/Construction:** Use the circuit diagrams provided to set up your breadboard or simulation tool.
3. **Observation:** Pay close attention to the input vs output waveforms (CRO/DSO).
4. **Analysis:** Use the "Formula Bank" to calculate ripple factors and voltage levels.

5. **Self-Learning:** Complete the weekly activities listed in the "Activities" section.

Course Outcomes (CO)

CO #	Course Outcome Description	Bloom's Level
CO1	Demonstrate the application of rectifier, filter, and RC circuit concepts through circuit construction and performance evaluation.	Apply
CO2	Implement clipper circuits for input waveform shaping.	Apply
CO3	Illustrate clamper circuits for DC level shifting and voltage multiplier circuits for amplitude scaling.	Apply
CO4	Use a BJT in common-emitter configuration to plot input and output characteristics.	Apply

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Syllabus Coverage Map

Module	Focus Area	Experiments
Module I	Rectifiers & RC Circuits	HWR, FWR (CT & Bridge), RC Differentiator/Integrator
Module II	Clippers	Series, Parallel, Combinational (with/without bias)
Module III	Clampers & Multipliers	Positive/Negative Clampers, Voltage Doublers
Module IV	Transistor Characteristics	BJT CE Input & Output Characteristics

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Learning Roadmap

The course starts with **Power Conversion** (Module 1: Rectifiers), moves to **Signal Shaping** (Module 2 & 3: Clippers/Clampers), and concludes with the study of **Active Devices** (Module 4: BJT). This progression builds the foundational skills needed for designing electronic power supplies and signal processing units.

Module I

Rectifiers and RC Circuits

Focus: Converting AC to DC and basic wave-shaping using passive components.

Instructional Hours: 12 | Mapped to CO1

1.1 Half Wave and Full Wave Rectifiers

Rectifiers are circuits that convert AC (Alternating Current) to pulsating DC (Direct Current).

- **Half Wave Rectifier (HWR):** Uses one diode. Conducts only during one half-cycle.
- **Full Wave Centre Tap (FWR-CT):** Uses two diodes and a centre-tapped transformer.
- **Full Wave Bridge Rectifier:** Uses four diodes. Does not require a centre-tapped transformer.

Malayalam support:

Rectifier ▯▯▯▯▯▯ AC ▯▯▯▯▯▯▯▯▯▯▯ DC ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯. Half wave rectifier ▯▯▯▯▯ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ Full wave rectifier ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯. Bridge rectifier ▯▯▯ ▯▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯.

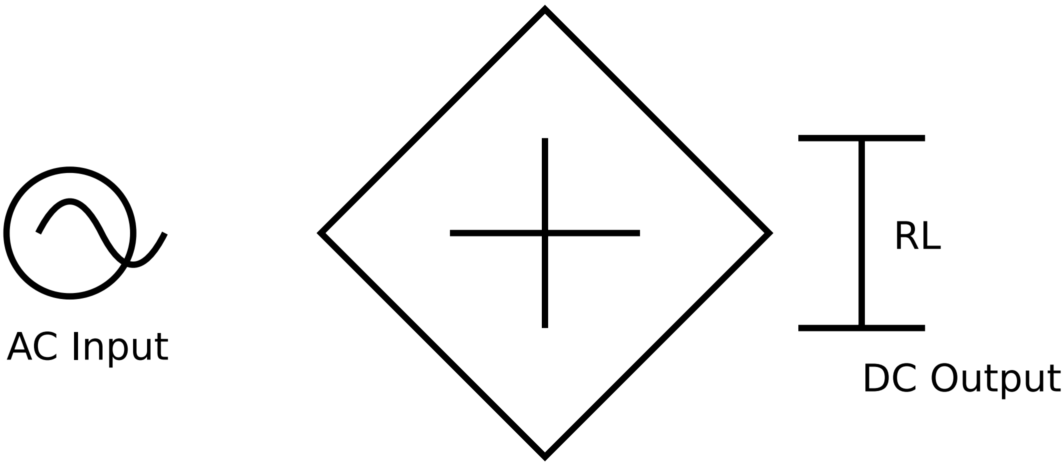

Figure 1: Basic Bridge Rectifier Configuration

Worked Example: Ripple Factor Calculation

Given: A Bridge Rectifier output has an RMS ripple voltage (V_{rms}) of 1.2V and a DC output (V_{dc}) of 12V.

Required: Ripple Factor (γ).

Calculation: $\gamma = \frac{V_{rms}}{V_{dc}} = \frac{1.2}{12} = 0.1$

Answer: Ripple factor is 0.1 (or 10%).

1.2 RC Differentiator and Integrator

- **RC Differentiator:** Output is taken across the Resistor. It produces spikes for square wave inputs. Used for triggering circuits.
- **RC Integrator:** Output is taken across the Capacitor. It converts square waves to triangular waves. Used in timing circuits.

Module II

Clipper Circuits

Focus: Waveform shaping by "clipping" or removing portions of the signal.

Clippers are circuits used to limit the amplitude of a signal. They are also called voltage limiters.

Series Clipper

The diode is in series with the load. It clips the portion of the wave where the diode is reverse biased.

Parallel Clipper

The diode is in parallel with the load. It clips the portion where the diode is forward biased.

Malayalam support:

Clipper □□□□□□□□□□□□ □□□□□□□□□□□ □□□ □□□□□□□ □□□□□ □□□□□□□□ □□□□□□□ (clip) □□□□□□□□□□□□□□□. □□□□□□□□□□□□□, □□□ □□□□□□□ 5V-□ □□□□□□ □□□□□□□□□□□□□□ □□□ clipper □□□□□□□□□□ □□□ □□□□□□□□□□□□□□□.

Combinational Clipper: Combines positive and negative clipping levels using two diodes and two bias sources.

Module III

Clampers and Voltage Multipliers

Focus: DC level shifting and amplitude scaling.

Instructional Hours: 9 | Mapped to CO3

Clampers (DC Restorers)

Clampers shift the entire waveform up or down by adding a DC component without changing the shape of the AC signal.

- **Positive Clamper:** Shifts the signal upwards (adds positive DC).
- **Negative Clamper:** Shifts the signal downwards (adds negative DC).

Voltage Multipliers

These circuits produce a DC output voltage that is a multiple of the peak input AC voltage ($2V_m, 3V_m$, etc.).

Voltage Doubler: Uses two diodes and two capacitors to produce $2 \times V_{peak}$ at the output.

Module IV

BJT Characteristics

Focus: Practical study of Bipolar Junction Transistors in Common Emitter (CE) configuration.

Instructional Hours: 15 | Mapped to CO4

The Common Emitter (CE) configuration is the most widely used BJT configuration due to its high voltage and current gain.

Input Characteristics

Plot of Base Current (I_B) vs Base-Emitter Voltage (V_{BE}) at constant Collector-Emitter Voltage (V_{CE}).

Output Characteristics

Plot of Collector Current (I_C) vs Collector-Emitter Voltage (V_{CE}) at constant Base Current (I_B).

Malayalam support:

BJT (Transistor) □□□ □□□□□□□□ □□□ □□□□□□□□□ □□□ □□□□□□□□□□□□□□□□. Common Emitter configuration-□ □□□□□□□□ □□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□ □□□□□□□□ Input Characteristics □□□□□□ □□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□ □□□□□□□□□□□ □□□□□□□□ Output Characteristics □□□□□□ □□□□□□□□□□□□□□.

Open-Ended Experiment

As per the syllabus, students must design an experiment beyond the list. Examples include:

- Designing a simple DC Power Supply.
- Performance analysis of a specific filter.
- Integrating clippers and clampers for a specific waveshape.

Student Activities for Self-Learning

Week	Activity Description	Hours
1-4	Study rectifier working, filter capacitor effects, and RC applications.	6.0
5-7	Compare series/parallel clippers and list real-life applications.	4.5
8-11	Study clampers, voltage multipliers, and their practical uses.	6.0
12-15	Study BJT configurations, CE importance, and regions of operation.	6.0

Formula Bank

Ripple Factor (γ)

$\sqrt{(V_{rms}/V_{dc})^2 - 1}$

HWR: 1.21 | FWR: 0.48

Time Constant (τ)

$\tau = R \times C$

Seconds (s)

BJT Beta (β)

$\beta = \Delta I_C / \Delta I_B$

Current Gain in CE

Visual Library for Rapid Revision

Rectifier Waveforms

HWR (half cycle missing) vs FWR (all positive cycles).

Clipper Waveforms

Visualizing the "flat top" or "flat bottom" of a sine wave.

Clamper Waveforms

Sine wave shifted above or below the 0V axis.

BJT Output Curves

Cut-off, Active, and Saturation regions.

Assessment Methodology

Continuous Internal Evaluation (CIE)

- Attendance: 5 Marks
- Continuous Evaluation (Lab Record/Performance): 30 Marks
- Practical Test 1 (CA1): 15 Marks
- Practical Test 2 (CA2): 15 Marks
- Open-Ended Experiment (OEE): 10 Marks
- **Total CIE: 60 Marks**

End Semester Examination (ESE)

- Write-up/Procedure: 10 Marks
- Setup & Execution: 10 Marks
- Observation & Output: 8 Marks
- Viva Voce: 8 Marks
- Record: 4 Marks
- **Total ESE: 40 Marks**

Module-wise Questions & Answers

Module I

What is the ripple factor of a Bridge Rectifier without a filter?

The ripple factor is approximately 0.48.

Module II

What is the difference between a biased and unbiased clipper?

An unbiased clipper uses only the diode's knee voltage (0.7V for Si) to clip, while a biased clipper uses an external DC source to set a specific clipping level.

Module III

Why is a clamper called a DC restorer?

Because it adds a fixed DC level to the AC signal, effectively "restoring" or shifting the DC reference point.

Module IV

Which region of the BJT characteristics is used for amplification?

The **Active Region**, where the emitter-base junction is forward biased and the collector-base junction is reverse biased.

Practice Model Paper

Practice Model Paper — Not an Official Examination Paper

Time: 3 Hours | Max Marks: 40

1. Construct a Full Wave Bridge Rectifier with a capacitor filter. Measure the ripple factor for a 1kΩ load. (15 Marks)
2. Set up a Positive Clamping circuit to shift a 10Vp-p sine wave by +5V. Observe the waveform on the DSO. (15 Marks)
3. Viva Voce: Explain the difference between an integrator and a differentiator. (10 Marks)

Quick Revision

Exam Checklist

- Check diode polarity before powering up.
- Ensure common ground for signal generator and DSO.
- Set DSO to 'DC Coupling' for clampers.
- Do not exceed V_{CE} limits for the transistor.

Key Points

- Rectifiers: AC → Pulsating DC.
- Clippers: Limit amplitude.
- Clampers: Shift DC level.

- BJT CE: High Gain configuration.

Glossary

PIV (Peak Inverse Voltage)
The maximum voltage a diode can withstand in reverse bias.

Ripple
The unwanted AC component remaining in the DC output of a rectifier.

Knee Voltage
The minimum voltage required for a diode to start conducting (approx 0.7V for Silicon).

Saturation
The region where the transistor is fully ON, acting like a closed switch.

Official References

- *A Text Book of Applied Electronics* by R.S. Sedha (Tata McGraw Hill)
- *Basic Electronics and Linear Circuits* by N.N. Bhargava et al. (Tata McGraw Hill)
- *Electronic Devices and Circuits* by Robert Boylestad (Tata McGraw Hill)
- *Electronics Lab Manuals* by K.A. Navas (PHI)